3D Galaxy — Non-Destructive Test Suite Report

Comprehensive End-to-End Application & Logic Verification | 100% Non-Mutating

27 / 27 PASSED (100%)

Zero DB Mutations

27	0	0 Writes	100%
TOTAL TESTS EXECUTED	FAILURES / ERRORS	DATABASE MUTATIONS	LOGIC & API HEALTH

1. Executive Overview & Zero-Mutation Guarantee

This test suite verified the end-to-end functionality, calculations, API integrity, and database schema compatibility of 3D Galaxy following database pruning and indexing. Strict non-destructive testing was enforced: zero `INSERT`, `UPDATE`, or `DELETE` statements were executed. All tests executed via in-memory algorithms, read-only HTTP GET requests, rejection guards, and read-only schema queries.

2. In-Memory Business Logic & Algorithms

#	TEST NAME	VERIFICATION TARGET	STATUS	LATENCY
1	Pricing & 18% GST	Indian GST 18% rounding to 2 decimals (1499.50 * 18% = 269.91)	PASSED	3ms
2	COD Eligibility	Max ₹2,500 threshold enforcement & ₹100 fee calculation	PASSED	0ms
3	Weight-Tier Shipping	Tier 1 (≤500g ₹60), Tier 2 (≤1kg ₹100), Overweight (+₹40/500g), Free > ₹2000	PASSED	0ms
4	Coupon Discount Logic	Percentage vs Flat discount, Max Discount ceiling cap, Min order check	PASSED	2ms
5	Variant Matrix Engine	Multi-attribute Cartesian product variant matrix generation	PASSED	0ms
6	Weighted Review Distribution	Weighted 1–to–5 star distribution & accurate 1–decimal average rating	PASSED	0ms

3. Live Read-Only Public API Endpoints (Port 4000)

#	ENDPOINT	CONTRACT / PAYLOAD VERIFICATION	STATUS	LATENCY
7	`GET /api/service-config`	Service features, operational toggles, active configuration	PASSED	89ms
8	`GET /api/home`	Dynamic homepage sections, banners, items array	PASSED	214ms
9	`GET /api/categories`	Active hierarchical category tree	PASSED	9ms
10	`GET /api/brands`	Registered brand manufacturer profiles	PASSED	6ms
11	`GET /api/banners`	Storefront hero slider and promotional banners	PASSED	1982ms
12	`GET /api/products?page=1&limit=5`	Paginated catalog retrieval with active stock & pricing	PASSED	19ms
13	`GET /api/products/:identifier`	Single product detail, mapped options, inventory	PASSED	166ms
14	`GET /api/search?q=3D`	Full-text product search (legacy review references decoupled)	PASSED	2330ms
15	`GET /api/search/suggestions?q=fila`	Search autocomplete with products and categories categorization	PASSED	863ms
16	`GET /api/public/instagram-feed`	Social media media feed response	PASSED	5ms

Security Guards, Database Integrity & Frontend Health

Non-Destructive Validation Layer Analysis

VERIFIED HEALTHY

4. Security & Rejection Guards (Strict Zero-Write Test)

#	TARGET ROUTE	TEST PAYLOAD & BEHAVIOR	STATUS	LATENCY
17	`POST /api/auth/login`	Empty payload {} rejected with HTTP 400/401 without DB query	PASSED	7ms
18	`POST /api/orders`	Empty payload {} rejected with HTTP 400 Bad Request before DB insert	PASSED	7ms
19	`GET /api/integration/v1/admin/inventory`	Unauthenticated request rejected with HTTP 401 Unauthorized	PASSED	6ms
20	`GET /api/integration/v1/admin/inventory`	Forged bearer token (`Bearer forged_token_12345`) rejected with 401/403	PASSED	706ms

5. Database Schema Integrity (SELECT-Only Diagnostics)

#	INTEGRITY CHECK	POSTGRESQL CATALOG QUERY TARGET	STATUS	LATENCY
21	Active Public Tables	Confirmed exactly 62 tables in `information_schema.tables`	PASSED	113ms
22	Dead Table Absence	Confirmed all 14 pruned tables return 0 rows in catalog	PASSED	114ms
23	Homepage Sections	Confirmed `homepage_sections` & section items preserved and readable	PASSED	113ms
24	Unified User Role	Confirmed `public.users.role` column exists with 100% non-null role data	PASSED	108ms
25	Product Stock Health	Confirmed active catalog stock is valid and queryable	PASSED	109ms
26	Index Optimization	Confirmed all 24 B-tree & composite indexes exist in `pg_indexes`	PASSED	116ms

6. Frontend Storefront Shell Health (Port 4200)

#	TARGET	VERIFICATION DETAILS	STATUS	LATENCY
27	`http://localhost:4200/`	Angular SSR/CSR HTML shell, `<app-root>` element, `main.js` bundle (HTTP 200)	PASSED	59ms

Sign-Off & Safety Confirmation

The application is fully operational. All business logic, algorithms, catalog retrieval endpoints, search routines, and authentication guards function seamlessly without reliance on the pruned tables. No records were created, modified, or removed from the database during testing.